8. Since Glauber's salt is efflorescent substance, it loses water of crystallization. These water molecules combine with anhydrous copper sulphate, to form hydrated copper sulphate which is blue in colour.

$CuSO_4 + 5H_2O \rightarrow CuSO_4.\ 5H_2O$

Colourless Blue

Hence, the correct option is (d).

9. Student C's observation is correct.

Hence, the correct option is (c).

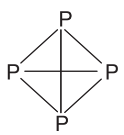
10. Butter is a gel that is colloid in which the dispersion medium is solid and the dispersed phase is liquid.

Hence, the correct option is (c).

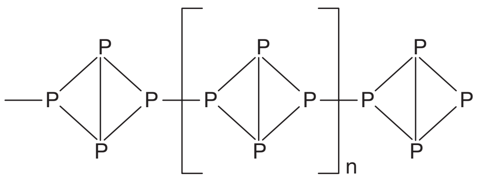
11. Since pig iron is brittle in nature, it cannot be used for making any article.

Hence, the correct option is (c).

12.



White phosphorus. Isolated tetrahedral p_4 units



Red phosphorus. Chain of tetrahedral P_4 units linked to each other by P–P bonds.

Hence, the correct option is (b).

13. $CuSO_4 + H_2S \rightarrow CuS \downarrow + H_2SO_4$. Copper sulphate Black precipitate

Hence, the correct option is (a).

14. $\underset{(aq)}{NaBr} + \underset{(aq)}{AgNO_3} \rightarrow \underset{\text{Pale yellow precipitate}}{AgBr \downarrow} + NaNO_3.$

$\underset{(aq)}{NaI} + \underset{(aq)}{AgNO_3} \rightarrow \underset{\text{Pale yellow precipitate}}{AgI \downarrow} + NaNO_3.$

Hence, the correct option is (d).

15. The property exploited is prevention of oxidation of tungsten.

Hence, the correct option is (c).

Assessment Test IV

1. Heat energy required

$Q = m\,s\,\Delta t,\ s = 4.2\ \text{J/g°C}$

$= 3 \times 4.2 \times 5$

$= 63\ \text{J g}^{-1}\ \text{C}^{-1}$

Hence, the correct option is (d).

2. Each carbon atom in graphite is attached to three other carbon atoms, forming a planar structure. Many such planar rings attach themselves to form a hexagonal-layered structure. Each carbon is attached to three other carbon atoms and one electron is left behind, which is responsible for the conductivity.

Hence, the correct option is (a).

3. According to Faraday's I law, $m \propto Q$

$i = 5$ amp, $t = 15$ min $\Rightarrow Q = 5 \times 15 \times 60 = 4500$ C

$i = 12$ amp, $t = 6$ min $\Rightarrow Q = 12 \times 6 \times 60 = 4320$ C

$i = 8$ amp, $t = 570$ s $\Rightarrow Q = 8 \times 570 = 4560$ C

$i = 14$ amp, $t = 330$ s $\Rightarrow Q = 14 \times 330 = 4620$ C

Hence, the correct option is (c).

4. Ore of Mg is Epsom salt; ore of Pb is anglesite; ore of Fe is limonite; ore of Al is cryolite; and ore of U is pitch blende.

Hence, the correct option is (c).

5. Removal of grease $\rightarrow NH_3$

Storage of canned food $\rightarrow N_2$

Manufacture of gunpowder $\rightarrow$ S

Preparation of poisonous gases $\rightarrow Cl_2$

Hence, the correct option is (b).

6. $b = \dfrac{W_{solute}}{a} \cdot 100$

$\therefore W_{solute} = \dfrac{ab}{100}\text{g}$

$\therefore V_{solution} = \dfrac{a}{x}\text{mL}$

$M = \dfrac{W_{solute}}{GMM} \times \dfrac{1000}{V}$, GMM of $Ca(OH)_2 = 74$

$$M = \frac{ab}{100 \times 74} \times \frac{1000\,x}{a}$$

$M = 0.135\ b\ x\ M$

Hence, the correct option is (b).

7. On cooling from 60°C to 25°C, B forms 180 – 140 = 40 g of precipitate and that of C forms 140 – 130 = 10 g of precipitate.

   Hence, the correct option is (d).

8. Hygroscopic substances which are used to remove moisture from the surroundings are called desiccating agents.

   Hence, the correct option is (b).

9. Addition of potash alum makes the clay particles, which are present in water, settle down. This refers to coagulation of clay particles by alum.

   Hence, the correct option is (c).

10. In foam, dispersion medium is liquid and dispersed phase is gas.

    Hence, the correct option is (a).

11. Wrought iron is the purest form of iron and is more malleable.

    Hence, the correct option is (a).

12. $X \rightarrow 2, 8, 5 \Rightarrow x$ is P; A is white phosphorous, B is red phosphorus. White phosphorous is less reactive than red phosphorus.

    Hence, the correct option is (d).

13. X is Mn

    $$MnSO_4 + H_2S \rightarrow \underset{\text{Flesh colour precipitate}}{MnS\downarrow} + H_2O.$$

    Hence, the correct option is (d).

14. $$2HI + HgCl_2 \rightarrow 2HCl + \underset{\text{Scarlet red}}{HgI_2\downarrow}$$

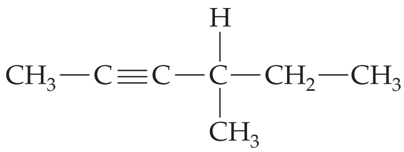
    Hence, the correct option is (a).

15. Neon is used in advertisement sign boards.

    Hence, the correct option is (d).

CHAPTER 7
Organic Chemistry

Assessment Test I

1. 

4–Methyl 2–hexyne

Total sigma bonds = $(3 \times 3) + (2 \times 1) + 7 = 18$

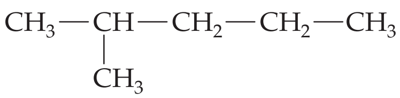
Total pi bonds = 2

Hence, the correct option is (a).

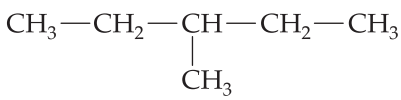
2. C_6H_{14} (Hexane) can show.

(a) $CH_3—CH_2—CH_2—CH_2—CH_2—CH_3$

n–Hexane

(b) 

2–methyl pentane

(c) 

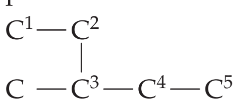
3–methyl, pentane

a, *b* and *a*, *c* are chain isomers.

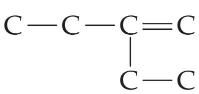
b, c are position isomers.

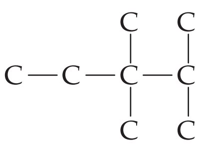
Hence, the correct option is (d).

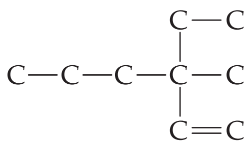
3. In 3-methyl 1-pentene, root word is pent and position of substituent is 3.

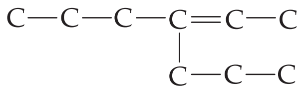
 → Root word is pent and position of substituent is 3.

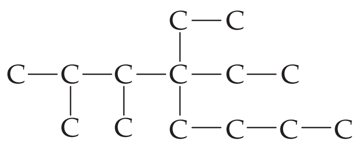
Hence, the correct option is (d).

4. (A)  Root word is but.

(B)  Root word is pent.

(C)  Root word is hex.

(D)  Root word is hept.

(E)  Root word is oct.

Hence, the correct option is (c).

5. (C) Petroleum gas (Below 30°C)

(B) Kerosene oil (150°C – 300°C)

(E) Fuel oil (300°C – 350°C)

(A) Lubricating oil (350°C – 400°C)

(D) Paraffin wax (Above 400°C)

Hence, the correct option is (d).

6. Ethene undergoes addition reaction with bromine water to give 1, 2-dibromo ethane. Hence, bromine water can be used to distinguish ethane and ethene.

Hence, the correct option is (a).

7. Both alkene and alkyne decolurise pink colour of alkaline $KMnO_4$ solution. Ethyne gets oxidized to oxalic acid.

Hence, the correct option is (c).

8. Green oil → Anthracene; Heavy oil → Phenol

Residue → Pitch; Light oil → Benzene

Hence, the correct option is (a).

9. If X is alkene; $12 \times 5 + 1 \times 10 = 70 \Rightarrow n = 5$

$\therefore$ X is C_5H_{10} and nearest neighbouring homologues are C_4H_8 and C_6H_{12}.

Hence, the correct option is (a).

10. $\underset{\text{Ethyl alcohol}}{C_2H_5{-}OH} \xrightarrow[350°C]{Al_2O_3} \underset{\substack{\text{Ethene}\\(A)}}{CH_2{=}CH_2} \xrightarrow{O_3/Zn,H_2O} \underset{\substack{\text{Formaldehyde}\\(B)}}{2HCHO}$

Hence, the correct option is (b).

11. $C_xH_y + (x + y/4)O_2 \longrightarrow xCO_2 + y/2H_2O$

$\therefore x = 2, y/2 = 2 \Rightarrow y = 4$

$\therefore$ X is C_2H_4 (ethene)

Ethene can be prepared by dehydration of ethyl alcohol in the presence of conc. H_2SO_4 at 180°C.

Hence, the correct option is (b).

12. Alkyne with least number of carbon atoms is ethyne.

$$\underset{\text{Ethyne}}{CH{\equiv}CH} \xrightarrow{Br_2} \underset{\text{1, 2–dibromo ethene}}{\underset{Br\quad\ \ Br}{CH{=}CH}} \xrightarrow{Br_2} \underset{\text{1, 1, 2, 2-tetrabromo ethane}}{\overset{Br\quad\ \ Br}{\underset{Br\quad\ \ Br}{CH{-}CH}}}$$

Hence, the correct option is (d).

13. Since chlorination of methane is a substitution reaction, the total number of bonds remains same.

Hence, the correct option is (a).

14. Saturated hydrocarbon with vapour density 8 is methane which can be prepared by decarboxylation of sodium acetate.

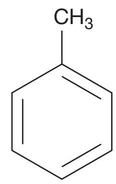
Hence, the correct option is (b).

15. Butanol and diethyl ether are functional isomers.

Hence, the correct option is (c).

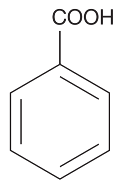
Assessment Test II

1. Toluene



$$\text{Total C – C bonds} = \underset{\text{sigma}}{\underset{\downarrow}{7}} + \underset{\text{pi}}{\underset{\downarrow}{3}} = 10$$

Benzoic acid



Benzoic acid

$$\text{Total C – C bonds} = \underset{\text{sigma}}{\underset{\downarrow}{7}} + \underset{\text{pi}}{\underset{\downarrow}{3}} = 10$$

Hence, the correct option is (d).

2. X is C_5H_{10} (2-methyle 1-butene)

(a) $CH_2{=}\underset{CH_3}{\underset{|}{C}}{-}CH_2{-}CH_3$

2–methyl 1–butene (X)

(b) $CH_2{=}CH{-}\underset{CH_3}{\underset{|}{CH}}{-}CH_3$

3–methyl 1–butene

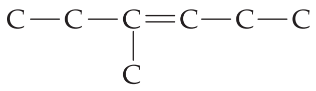
(c) $CH_3{-}C{=}\underset{CH_3}{\underset{|}{CH}}{-}CH_3$

2–methyl 2 – butene

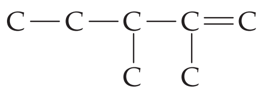
(d) $CH_2{=}CH{-}CH_2{-}CH_2{-}CH_3$

1–pentene is the chain isomer of X.

Hence, the correct option is (d).

3. In 

Position of substituents is 3 and position of double bond is 3. Sum = 3 + 3 = 6

In 

Positions of two substituents are 2 and 3. Position of double bond is 1.

$\therefore$ Sum = 2 + 3 + 1 = 6

Hence, the correct option is (b).

4. (A) 2 – ethyl pentane $\rightarrow$ Total no. of carbon atoms is 7.

(B) 4 – methyl 1 – heptene $\rightarrow$ Total no. of carbon atoms is 8.

(C) 3, 3 – diethyl pentane $\rightarrow$ Total no. of carbon atoms is 9.

(D) 3, 4, 5 – trimethyl heptane $\rightarrow$ Total no. of carbon atoms is 10.

(E) 5 – ethyl nonene $\rightarrow$ Total no. of carbon atoms is 11.

Hence, the correct option is (b).

5. (A) Fuel oil $\rightarrow C_{16} - C_{20}$

(B) Paraffin wax $\rightarrow$ Above C_{24}

(C) Crude naphtha $\rightarrow C_5 - C_{10}$

(D) Kerosene oil $\rightarrow C_{10} - C_{16}$

(E) Lubricating oil $\rightarrow C_{20} - C_{24}$

Hence, the correct option is (b).

6. Ammoniacal silver nitrate solution can be used to distinguish alkene and ethyne or 1 – alkyne. Ethyne or 1 – alkyne undergoes substitution reaction with ammoniacal silver nitrate solution.

Hence, the correct option is (c).

7. Saturated and unsaturated [alkene and alkyne) hydrocarbons can be distinguished by alkaline $KMnO_4$ solution. Alkaline $KMnO_4$ solution is a good oxidizing agent.

Hence, the correct option is (d).

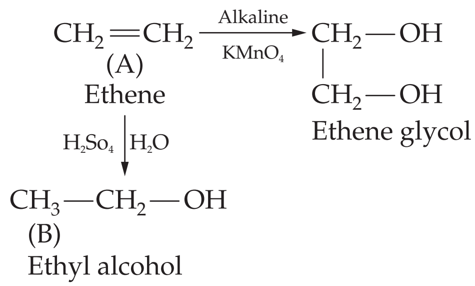
8. Middle oil $\rightarrow$ 170°C – 230°C ; Heavy oil $\rightarrow$ 230°C – 270°C

Green oil $\rightarrow$ 270°C – 400°C ; Residue $\rightarrow$ above 400°C

Hence, the correct option is (d).

9. The molecular weight difference between adjacent homologues is 14. The difference in the molecular weights is 56. The molecular formulae are C_4H_{10} and C_8H_{10}.

Hence, the correct option is (d).

10. 

Hence, the correct option is (c).

11.
$$\underset{\text{Hexane (X)}}{CH_3—CH_2—CH_2—CH_2—CH_2—CH_3} \xrightarrow{\text{Heat}} \underset{\text{Butane(Y)}}{CH_3—CH_2—CH_2—CH_3} + \underset{\text{Ethene (Z)}}{CH_2{=}CH_2}$$

Ethene can be prepared by dehydration of ethyl alcohol in the presence of Al_2O_3 at 170°C.

Hence, the correct option is (c).

12. In ethyne the ratio of number of carbon atoms to hydrogen atoms is 1:1. Ethyne on ozonolysis gives glyoxal.

Hence, the correct option is (b).

13. In hydrogenation of alkene, the total number of bonds increases by one.

Hence, the correct option is (b).

14. Hydrocarbon X is CH_4. CH_4 on pyrolysis gives C and H_2.

Hence, the correct option is (a).

15. Alcohol X is butanol which can have two position isomers.

Hence, the correct option is (b).

Assessment Test III

1. Since C_4H_8 is an unsaturated hydrocarbon that is alkene, it decolourises bromine water.

   Hence, the correct option is (a).

2. The major component of natural gas is methane, which, on pyrolysis, gives hydrogen gas. The hydrogen so obtained can be used in the synthesis of ammonia which is the most important component of nitrogenous fertilizers.

   Hence, the correct option is (a).

3. (a) Root word is Hex. (b) Root word is but.

   (c) Root word is Hept. (d) Root word is pent.

   $\therefore$ Order is BDAC.

   Hence, the correct option is (a).

4. Ethane + water $\rightarrow$ Ethyl alcohol $\Rightarrow$ Addition of water

   Ethene $\xrightarrow[H_2O]{O_3, Zn}$ Formaldehyde $\Rightarrow$ Ozonolysis

   Ethene $\xrightarrow{[O].KMNO_4}$ Ethylene glycol $\Rightarrow$ Oxidation

   Ethene + $O_2 \rightarrow CO_2 + H_2O \rightarrow$ Combustion

   Hence, the correct option is (c).

5. The correct matching is

   (i) $\rightarrow$ (B); (ii) $\rightarrow$ (C); (iii) $\rightarrow$ (D); (iv) $\rightarrow$ (A)

   Hence, the correct option is (d).

6. GMM $= 2 \times 36 = 72$

   X may be alkane/alkene/alkyne

   If X is alkane; $C_n\,H_{2n}+_2$, $n = 1, 2, 3, 4\ldots$

   $\therefore$ GMM $= 12n + 2n + 2 = 72$

   $\Rightarrow 14n = 70$

   $\Rightarrow n = 5$

   $\therefore$ X is $C_5\,H_{12}$.

$$\left.\begin{array}{l} CH_3-CH_2-CH_2-CH_2-CH_3 \\ n\text{-pentane} \\ CH_3-\underset{\underset{\displaystyle CH_3}{|}}{CH}-CH_2-CH_3 \\ \text{Isopentane} \\ CH_3-\overset{\overset{\displaystyle CH_3}{|}}{\underset{\underset{\displaystyle CH_3}{|}}{C}}-CH_3 \\ \text{Neo pentane} \end{array}\right\} \begin{array}{l}\text{Chain} \\ \text{isomers} \\ \text{of } C_5H_{12}\end{array}$$

   Hence, the correct option is (d).

7. $$\overset{1}{CH_3}-\underset{\underset{\displaystyle CH_3}{|}}{\overset{2}{CH}}-\overset{3}{CH_2}-\overset{4}{CH_2}-\overset{5}{CH_3}$$

   2–methyl pentane

   $$\overset{1}{CH_3}-\overset{2}{CH_2}-\underset{\underset{\displaystyle CH_3}{|}}{\overset{3}{CH}}-\overset{4}{CH_2}-\overset{5}{CH_3}$$

   3–methyl pentane

   Hence, the correct option is (a).

8. C_4H_8 is cyclobutane (or) butene.

   C_5H_8 is pentyne.

   C_6H_{14} is hexane.

   C_7H_{16} is heptane.

   Hence, the correct option is (a).

9. $$CaC_2 + H_2O \rightarrow \underset{\substack{(X) \\ \text{Ethyne}}}{C_2H_2} + Ca(OH)_2$$

   $$C_2H_2 + \frac{5}{2}O_2 \rightarrow 2CO_2 + H_2O$$

   $\therefore$ Number of moles of O_2 is $5/2 = 2.5$.

   Hence, the correct option is (a).

10. C_3H_8 is alkane, that is, propane.

    C_4H_6 is alkyne, that is, butyne.

    Hence, the correct option is (d).

11. Sunlight is required to break chlorine molecules into chlorine free radicals. These chlorine free radicals get involved in further reactions.

Hence, the correct option is (a).

12. $CH_3I + [H] \xrightarrow[C_2H_5-OH]{Zn-Cu} \underset{(X)}{CH_4} + HI$

$CH_4 + H_2O \xrightarrow[750°C]{Al_2O_3} \underset{(Y)}{CO} + \underset{(Z)}{3H_2}$

Hence, the correct option is (a).

13. $2CHI_3 + 6Ag \rightarrow \underset{\text{Ethyne}}{\underset{(X)}{CH \equiv CH}} + 6AgI$

Ethyne on polymerization gives benzene.

Hence, the correct option is (d).

14. The major component of light oil is benzene.

Hence, the correct option is (c).

15. $C_xH_y + \left(x + \frac{y}{4}\right)O_2 \rightarrow x.\,CO_2 + \frac{y}{2}H_2O$

Given that $\frac{x}{\left(\frac{y}{2}\right)} = 4:3$

$\Rightarrow \frac{2x}{y} = \frac{4}{3}$

$\Rightarrow \frac{x}{y} = \frac{4}{6}$

$\therefore$ X is C_4H_6

→ $CH \equiv C-CH_2-CH_3$, 1-butyne

→ $CH_3-C \equiv C-CH_3$, 2-butyne

Hence, the correct option is (b).

Assessment Test IV

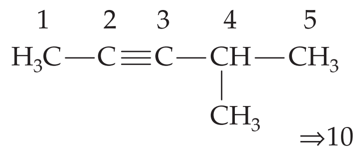
1. Vegetable oil + $H_2 \xrightarrow[47.3k,\,5atm]{Ni}$ Vanaspati

Since vanaspati is a hydrogenated oil, the unsaturated positions get to saturate. As a result, oxidation is prevented.

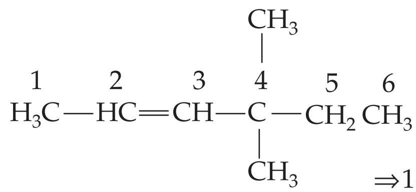
Hence, the correct option is (c).

2. The major component of CNG is methane and that of LPG are propane and butane. Methane, on combustion, produces the least amount of CO_2 when compared to propane and butane. Moreover, propane and butane show more knocking property which reduces fuel efficiency.

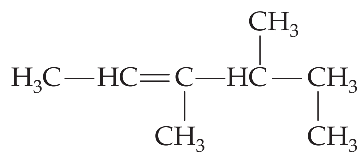
Hence, the correct option is (a).

3. 

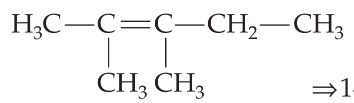
$\Rightarrow$10 hydrogen atoms



$\Rightarrow$16 hydrogen atoms



$\Rightarrow$18 hydrogen atoms



$\Rightarrow$14 hydrogen atoms

Hence, the correct option is (a).

4. Ethyne $\xrightarrow[H_2O]{O_3,Zn}$ Glyoxal $\Rightarrow$ Ozonolysis

Ethyne $\xrightarrow[KMnO_4]{[O]}$ Oxalic acid $\Rightarrow$ Oxidation

Ethyne $\xrightarrow[\text{Red hot Cu}]{300°C}$ Benzene $\Rightarrow$ Polymerisation

Ethyne $\xrightarrow[H_2SO_4 + HgSO_4]{H_2O}$ Acetaldehyde $\Rightarrow$ Addition of water

Hence, the correct option is (b).

5. The correct matching is

(i) → (B); (ii) → (D); (iii) → (A); (iv) → (C)

Hence, the correct option is (b).

6. $C_7H_{12} \Rightarrow CH{\equiv}C{-}CH_2{-}CH_2{-}CH_2{-}CH_2{-}CH_3$

1-heptyne

$$CH{\equiv}C{-}\underset{\overset{|}{CH_3}}{CH}{-}CH_2{-}CH_2{-}CH_3$$

3-methyl 1-hexyne

$$CH{\equiv}C{-}\underset{\overset{|}{CH_3}}{\overset{\underset{|}{CH_3}}{C}}{-}CH_2{-}CH_3$$

3, 3-dimethyl 1-pentyne

Hence, the correct option is (b).

7.

$H_2C{=}\underset{\overset{\vert}{CH_3}}{C}{-}CH_2{-}CH_2{-}CH_3$ 2-methyl 1-pentene	$CH_2{=}CH{-}\underset{\overset{\vert}{CH_3}}{CH}{-}CH_2{-}CH_3$ 3-methyl 1-pentene
$H_3C{-}\underset{\overset{\vert}{CH_3}}{C}{=}CH{-}CH_2CH_3$ 2-methyl 2-pentene	$H_3C{-}HC{=}\underset{\overset{\vert}{CH_3}}{C}{-}CH_2CH_3$ 3-methyl 2-pentene
$CH_2{=}CH{-}CH_2{-}\underset{\overset{\vert}{CH_3}}{CH}{-}CH_3$ 4-methyl 1-pentene	$H_3C{-}HC{=}CH{-}\underset{\overset{\vert}{CH_3}}{CH}{-}CH_3$ 4-methyl 2-pentene

Hence, the correct option is (d).

8. C_5H_8 is alkyne with one triple bond.

Hence, the correct option is (c).

9. GMM of X is $2 \times 13 = 26$.

Since X is unsaturated hydrocarbon, it is either alkene or alkyne.

If X is alkene C_nH_{2n}, $n = 2, 3, 4\ldots$

$GMM = 12n + 2n = 26$

$\Rightarrow 14n = 26$

$\Rightarrow n = \dfrac{26}{14}$, n could have only whole number

$\Rightarrow x$ is alkyne $C_n\,H_{2n-2}$; $n = 2, 3, \ldots$

$GMM = 12n + 2n - 2 = 26$

$\Rightarrow 14n = 28$

$\Rightarrow n = 2$

$\therefore$ X is C_2H_2(ethyne)

$$CH{\equiv}CH + 2Br_2 \longrightarrow Br{-}\underset{}{CH}{-}\overset{\overset{Br}{|}}{CH}{-}\overset{\overset{Br}{|}}{Br}$$

1, 1, 2, 2 tetra bromo ethane

Hence, the correct option is (d).

10. C_5H_8 and C_6H_{10} are homologues.

Hence, the correct option is (d).

11. $CH_4 + Cl_2 \xrightarrow{h\upsilon} CH_3Cl + HCl$

(Excess)

Hence, the correct option is (b).

12. $CH_3COONa \xrightarrow{NaOH+CaO} \underset{(X)}{CH_4} + Na_2CO_3$

$$\underset{(Y)}{4CO} + \underset{(Z)}{2H_2O} \xrightarrow[250°C]{NiCO_3} \underset{(X)}{CH_4} + CO_2$$

Hence, the correct option is (b).

13. $HC{\equiv}CH + 2AgNO_3 + 2NH_4OH$

$\downarrow$

$AgC{\equiv}CAg + 2NH_4NO_3 + 2H_2O$

silver acetylide white precipitate

$HC{\equiv}CH + 2CuCl \longrightarrow CuC{\equiv}CCu + 2HCl$

Copper acetylide Red precipitate

Hence, the correct option is (c).

14. The range of temperature is 170°C–270°C.

Hence, the correct option is (b).

15. $C_xH_y\left(x + \dfrac{y}{4}\right)O_2 \rightarrow x\,CO_2 + \dfrac{y}{2}H_2O$

Given that $x: \dfrac{y}{2} = 10 : 8$

$$\Rightarrow \frac{x}{\left(\frac{y}{2}\right)} = \frac{10}{8} \Rightarrow \frac{2x}{y} = \frac{10}{8}$$

$$\Rightarrow \frac{x}{y} = \frac{5}{8}$$

$\therefore$ X is C_5H_8.

$$\therefore \%\text{ of C} = \frac{5\times12\times100}{(5\times12)+8} = \frac{60\times100}{68} = 88.2\%$$

Hence, the correct option is (b).

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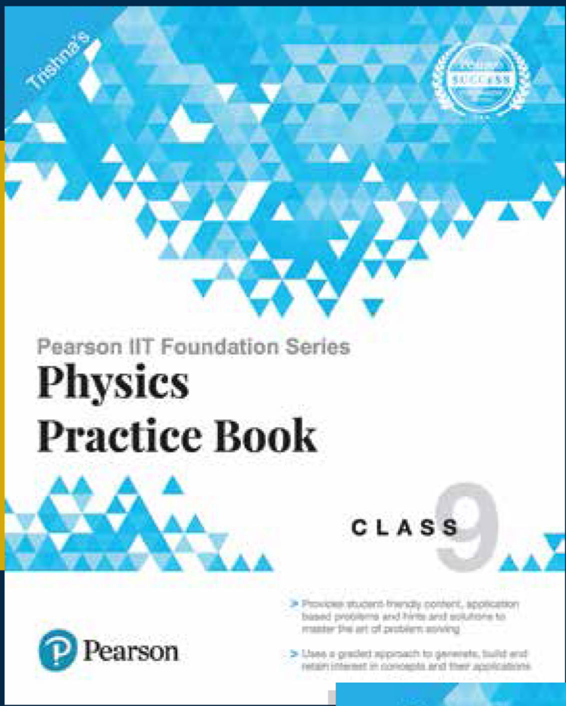

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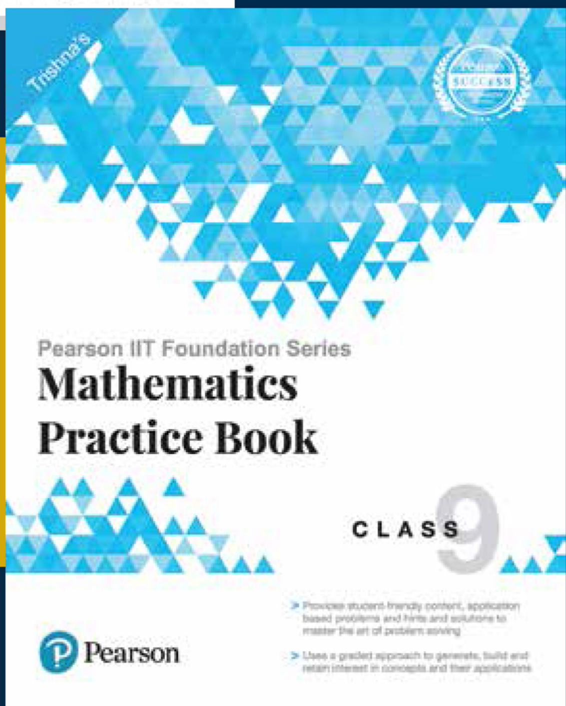

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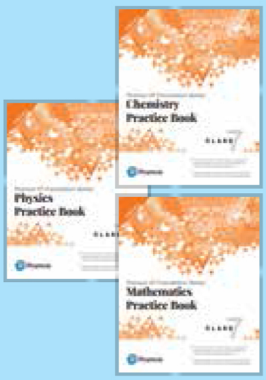

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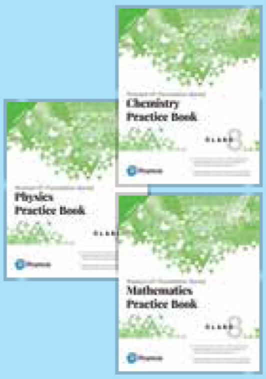

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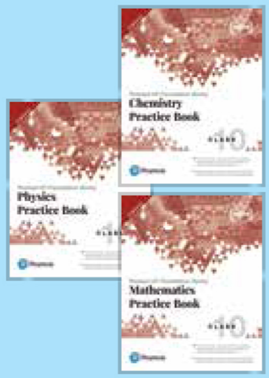

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